

CS445 - Final Presentation

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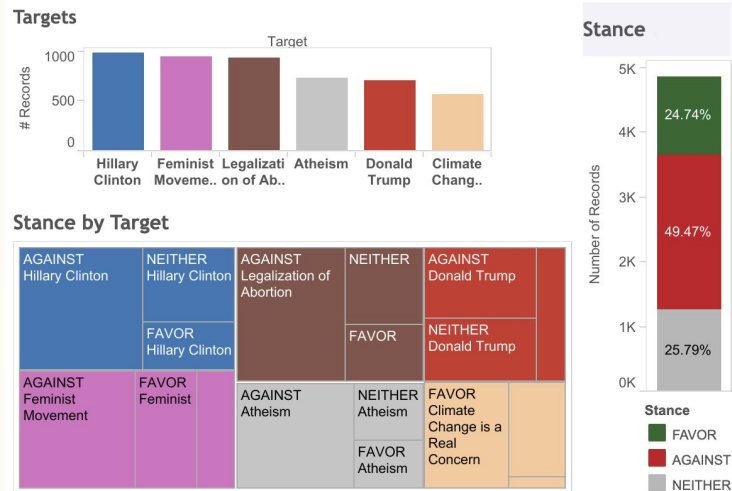
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Tweet Preprocessing

- Sem-Eval 2016 Task 6 Dataset, which consists of tweets with their stances ("FAVOR", "AGAINST", "NEITHER") against targets like "Abortion" and "Atheism"

- Converted to lowercase
- Special Character removal except for '-', '_', and '#'
- Extra spaces were removed



- Tweets with stance target 'Donald Trump' was removed from the test data as they were not included in the training data set

Recurrent Neural Network

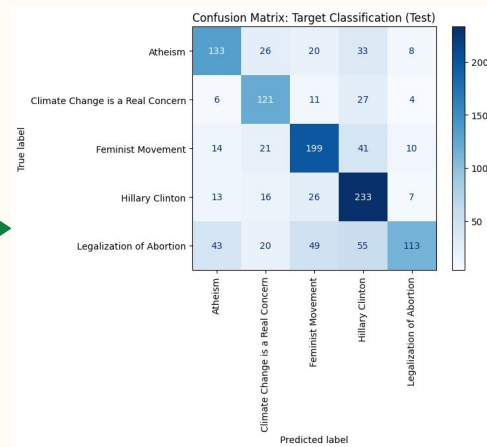
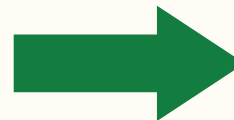
Target Classification (RNN)

- **Embedding Layer:** Vocabulary size of 10,000 and embedding dimension of 128.
- **LSTM Layer:** 128 units.
- **Dense Layers:** A ReLU-activated layer with 128 units and a softmax output layer.
- **Loss:** Sparse categorical cross entropy is used.
- **Optimizer:** Adam optimizer with 0.001 learning rate.
- **Epochs:** 20 epochs
- **Batch Size:** 32

Target Classification

Macro Precision [Test]:
0.6553412555822804

Macro Recall [Test]:
0.6424338673974986



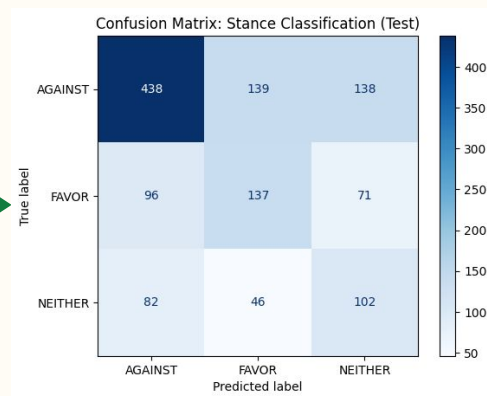
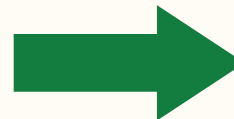
Stance Classification (RNN)

- **Embedding Layer:** Vocabulary size of 10,000 and embedding dimension of 128.
- **LSTM Layer:** 128 units.
- **Concatenation:** Combining the LSTM output with the target one-hot vector.
- **Dense Layers:** ReLU-activated hidden layer with 64 units and a softmax output layer.
- **Loss:** Sparse categorical cross entropy.
- **Optimizer:** Adam optimizer with a learning rate of 0.008.
- **Epochs:** 8.
- **Batch Size:** 32.

Stance Classification

Macro Precision [Test]:
0.4965155282467857

Macro Recall [Test]:
0.5095016349020925



Paper Reproduction and BERT

The model in Mohammed Et al.'s 2017 paper "Stance and Sentiment in Tweets" was replicated which consisted of:

1,2,3 word ngrams; 2, 3, 4, 5 character ngrams; presence of target and Word2Vec embeddings features were used with a linear SVM.

Mohammed et al.'s result was improved upon when BERT embeddings were used instead of Word2Vec embeddings

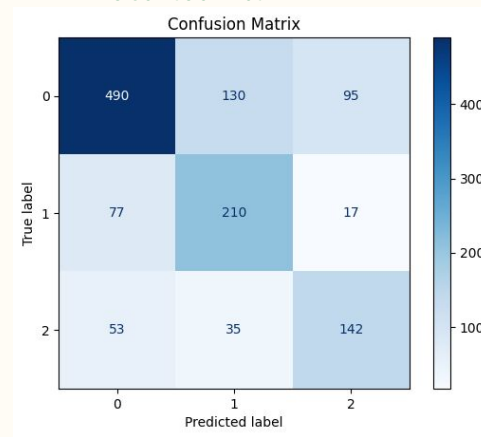
The BERT features also lead to greater performance with other models like Random Forest and SGDClassifier than the paper implementation.

The results were further improved when a BERT classifier was used instead of just the BERT embeddings

A multi classifier BERT model was fine tuned on the training data for 10 epochs with 8 batch size.

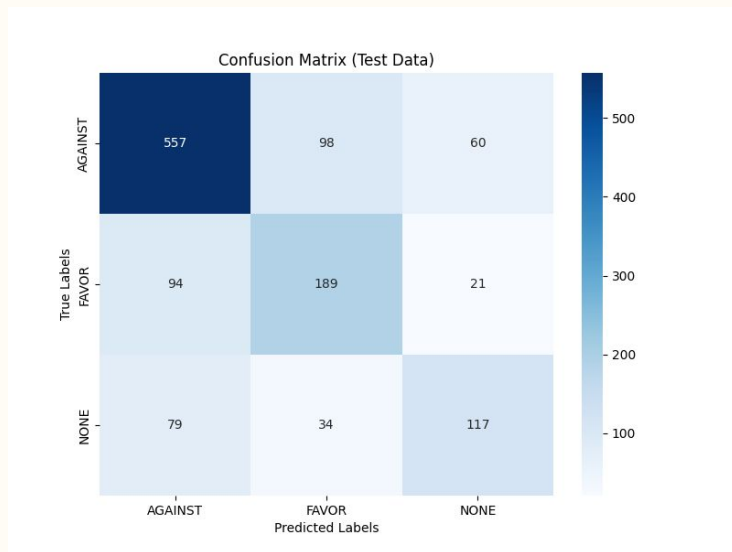
<i>Model</i>	<i>Accuracy</i>	<i>Micro F1</i>	<i>Macro F1</i>	<i>Macro Precision</i>	<i>Macro Recall</i>
<i>Paper's Results (Mohammed et al. 2017)</i>		<i>0.703</i>	<i>0.590</i>		
<i>Paper Reproduced (SVM + Word2Vec + ngrams + Target)</i>	<i>0.63</i>	<i>0.66</i>	<i>0.64</i>	<i>0.58</i>	<i>0.60</i>
<i>SVM + BERT + ngrams + Target</i>	<i>0.64</i>	<i>0.67</i>	<i>0.65</i>	<i>0.60</i>	<i>0.62</i>
<i>SGD + BERT + ngrams + Target</i>	<i>0.64</i>	<i>0.66</i>	<i>0.63</i>	<i>0.60</i>	<i>0.63</i>
<i>Random Forest + BERT + ngrams + Target</i>	<i>0.67</i>	<i>0.71</i>	<i>0.61</i>	<i>0.70</i>	<i>0.53</i>
<i>Fine Tuned BERT</i>	<i>0.67</i>	<i>0.67</i>	<i>0.64</i>	<i>0.64</i>	<i>0.66</i>

BERT's Confusion Matrix



*Where F1 scores are only based on the "FAVOR" and "AGAINST" classes like the original paper

Feedforward Neural Networks



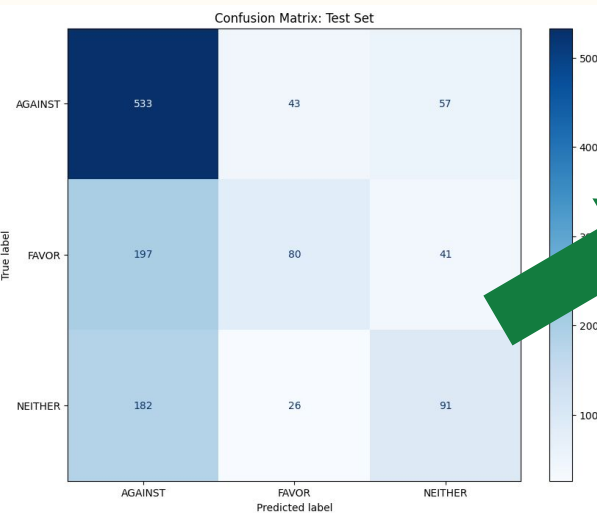
Confusion Matrix for Test Data

- **Optimizer:** Adam optimizer (learning rate = 0.0001)
- **Dropout rate:** 0.3
- **Epochs:** 10 **Batch Size:** 32
- **Target F1-Score (Test):** 0.69
- **Target Macro Precision (Test):** 0.65
- **Target Macro Recall (Test):** 0.64
- **Stance F1-Score (Test):** 0.64
- **Stance Macro Precision (Test):** 0.65
- **Stance Macro Recall (Test):** 0.64

Naive Bayes with BERT Word Embeddings

Model 1: Naive Bayes + BERT

Stance detection was performed with the Naive Bayes model by combining all target categories.: This model was especially successful in the AGAINST class, but showed low performance in the FAVOR and NEITHER classes.



Model 2: Category-Specific Naive Bayes + BERT

Separate Naive Bayes models were created for each target category and category-specific contexts were better captured. AGAINST class showed high performance in most categories. It gave better results compared to Model 1.

Training model for category: Feminist Movement

Validation Set Evaluation for category: Feminist Movement

	precision	recall	f1-score	support
AGAINST	0.540984	0.733333	0.622642	45.000000
FAVOR	0.166667	0.033333	0.055556	30.000000
NEITHER	0.315789	0.545455	0.400000	11.000000
accuracy	0.465116	0.465116	0.465116	
macro avg	0.341147	0.437374	0.359399	86.000000
weighted avg	0.381604	0.465116	0.396343	86.000000

Test Set Evaluation for category: Feminist Movement

	precision	recall	f1-score	support
AGAINST	0.664835	0.765823	0.711765	158.000000
FAVOR	0.366667	0.144737	0.207547	76.000000
NEITHER	0.413333	0.584906	0.484375	53.000000
accuracy	0.567944	0.567944	0.567944	
macro avg	0.481612	0.498488	0.467896	287.000000
weighted avg	0.539433	0.567944	0.536252	287.000000

Model 3: Category + Stance (Logistic Regression + Naive Bayes)

A two-stage process was implemented. In the first stage, category prediction was made using TF-IDF based Logistic Regression. In the second stage, separate stance detection models were created for each category.

